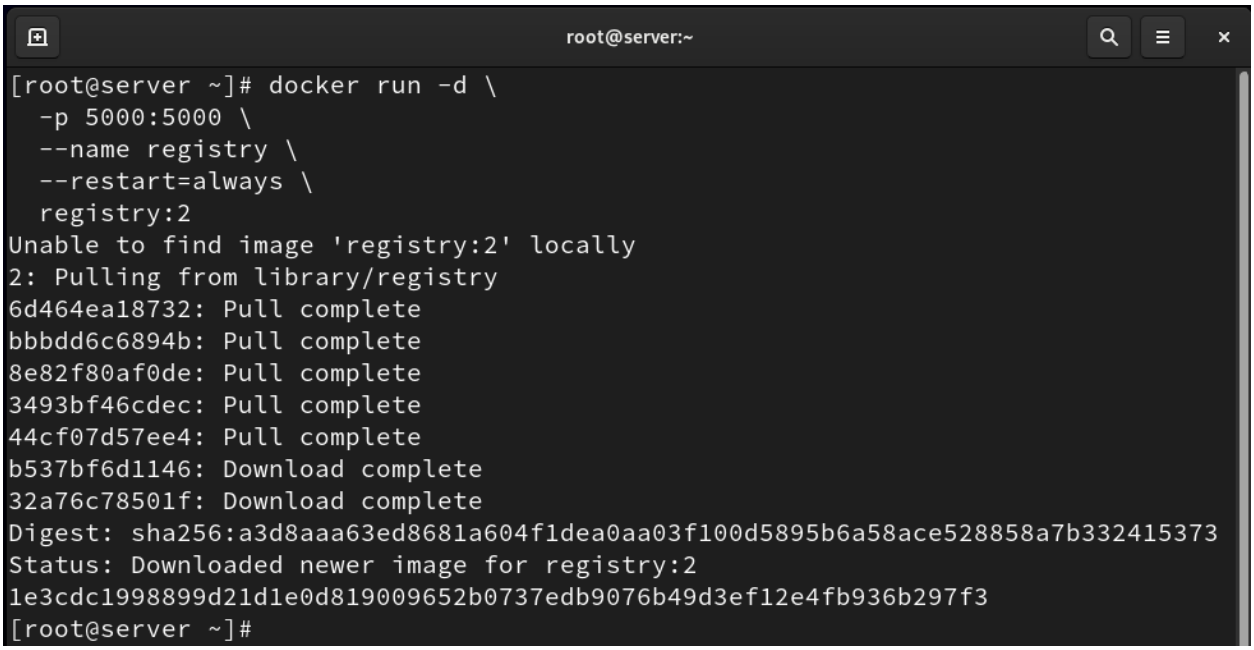
Lab 3

Part1

Step 1: Run an Insecure Docker Registry

```
docker run -d \
  -p 5000:5000 \
  --name registry \
  --restart=always \
  registry:2
```



Step 2: Configure Docker as Insecure Registry

sudo vi /etc/docker/daemon.json

Add:

```
{
  "insecure-registries": ["localhost:5000"]
}
```

sudo systemctl restart docker

```
root@server:~ — sudo vi /etc/docker/daemon.json
{
  "insecure-registries": ["localhost:5000"]
}
~
~
```

Step 3: Create Nginx Image Based on alpine

```
[root@server ~]# mkdir nginx-image
[root@server ~]# cd nginx-image
[root@server nginx-image]# vi Dockerfile
[root@server nginx-image]# cat Dockerfile
FROM alpine:latest

RUN apk add --no-cache nginx

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
[root@server nginx-image]#
[root@server nginx-image]#
```

Step 4: Build Image

docker build -t localhost:5000/nginx-alpine:latest .

```
[root@server nginx-image]# docker build -t localhost:5000/nginx-alpine:latest .
[+] Building 11.6s (6/6) FINISHED                                docker:default
 => [internal] load build definition from Dockerfile                        0.1s
 => => transferring dockerfile: 193B                                        0.0s
 => [internal] load metadata for docker.io/library/alpine:latest            0.3s
 => [internal] load .dockerignore                                           0.0s
 => => transferring context: 2B                                             0.0s
 => [1/2] FROM docker.io/library/alpine:latest@sha256:25109184c71bdad752c   0.0s
 => => resolve docker.io/library/alpine:latest@sha256:25109184c71bdad752c   0.0s
 => [2/2] RUN apk add --no-cache nginx                                     10.9s
 => exporting to image                                                      0.2s
 => => exporting layers                                                     0.1s
 => => exporting manifest sha256:d5d3122b365f61c4130fe35970088bd0a4e9994c   0.0s
 => => exporting config sha256:e635c3cc224f0d522434b3452f3c2ca06873995584   0.0s
 => => exporting attestation manifest sha256:797165d743d98144a23fddb6c8d0   0.0s
 => => exporting manifest list sha256:49f8d0e4a224f806664ed4ab243e8dd49e6   0.0s
 => => naming to localhost:5000/nginx-alpine:latest                         0.0s
 => => unpacking to localhost:5000/nginx-alpine:latest                      0.0s
[root@server nginx-image]#
```

Step 5: Push Image to Private Registry

docker push localhost:5000/nginx-alpine:latest

```
[root@server nginx-image]# docker push localhost:5000/nginx-alpine:latest
The push refers to repository [localhost:5000/nginx-alpine]
75c9bc4c8bfa: Pushed
6b7049f4a75e: Pushed
589002ba0eae: Pushed
latest: digest: sha256:49f8d0e4a224f806664ed4ab243e8dd49e666ad7e275fdbddf65de3c0
64a4a3f size: 855
[root@server nginx-image]#
```

Step 6: Verify that Image Uploaded to private Repo

curl http://localhost:5000/v2/_catalog

```
[root@server nginx-image]#
[root@server nginx-image]# curl http://localhost:5000/v2/_catalog
{"repositories":["nginx-alpine"]}
```

Part2

Step1: Create docker-compose.yml

mkdir wordpress-lab
cd wordpress-lab
vi docker-compose.yml

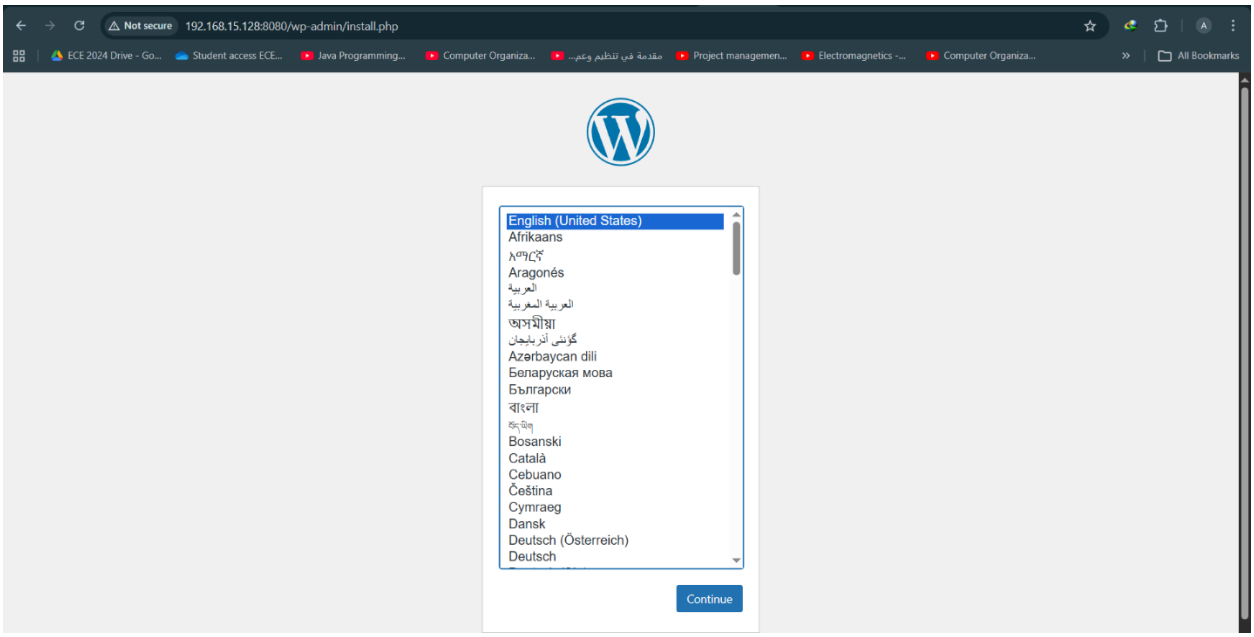
```yaml
version: '3.8'

services:

  db:
    image: mysql:5.7
    container_name: mysql_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpassword
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppassword
    volumes:
      - /root/mysql-data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress_app
    restart: always
    ports:
      - "8080:80"
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppassword
      WORDPRESS_DB_NAME: wordpress
    depends_on:
      - db
```

Step2: Run Compose

docker compose up -d

```
[root@server wordpress-lab]# docker compose up -d
WARN[0000] /root/wordpress-lab/docker-compose.yml: the attribute `version` is
obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 6/38
 ⠸ Image wordpress:latest [⣿⣿⣿
```

Step3: Verify



Part3

Step 1: Push Lab 2 Image to Private Registry

```
[root@server ~]# docker tag ahmedamer10/iti-flask-lab2:latest localhost:5000/iti-flask-lab2:latest
[root@server ~]# docker push localhost:5000/iti-flask-lab2:latest
The push refers to repository [localhost:5000/iti-flask-lab2]
27711589a568: Pushed
2d35ebdb57d9: Pushed
c9008197b8dd: Pushed
34b596050a95: Pushed
9806928fa0da: Pushed
cde5eaaf9876: Pushed
b410233c9818: Pushed
28264e704d4d: Pushed
latest: digest: sha256:d8d49ae757644ae2c47dec2858f71b8b2dd73bc6bd7f0aa4060fe61e4c4da63e size: 856
[root@server ~]#
```

Make sure that the image uploaded to the private repository

```
[root@server ~]# curl http://localhost:5000/v2/_catalog
{"repositories":["iti-flask-lab2","nginx-alpine"]}
[root@server ~]#
```

Step 2: Create docker-compose for Flask + Nginx

mkdir flask-lab3
cd flask-lab3
vi docker-compose.yml

```yaml
version: '3.8'

services:

  flask:
    image: localhost:5000/iti-flask-lab2:latest
    container_name: flask_app
    restart: always
    expose:
      - "5000"

  nginx:
    image: nginx:latest
    container_name: nginx_proxy
    restart: always
    ports:
      - "80:80"
    volumes:
      - ./default.conf:/etc/nginx/conf.d/default.conf
    depends_on:
      - flask
~
```

Step 3: Create Nginx Reverse Proxy Config

vi default.conf

```
server {
    listen 80;

    location / {
        proxy_pass http://flask:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
~
~
```

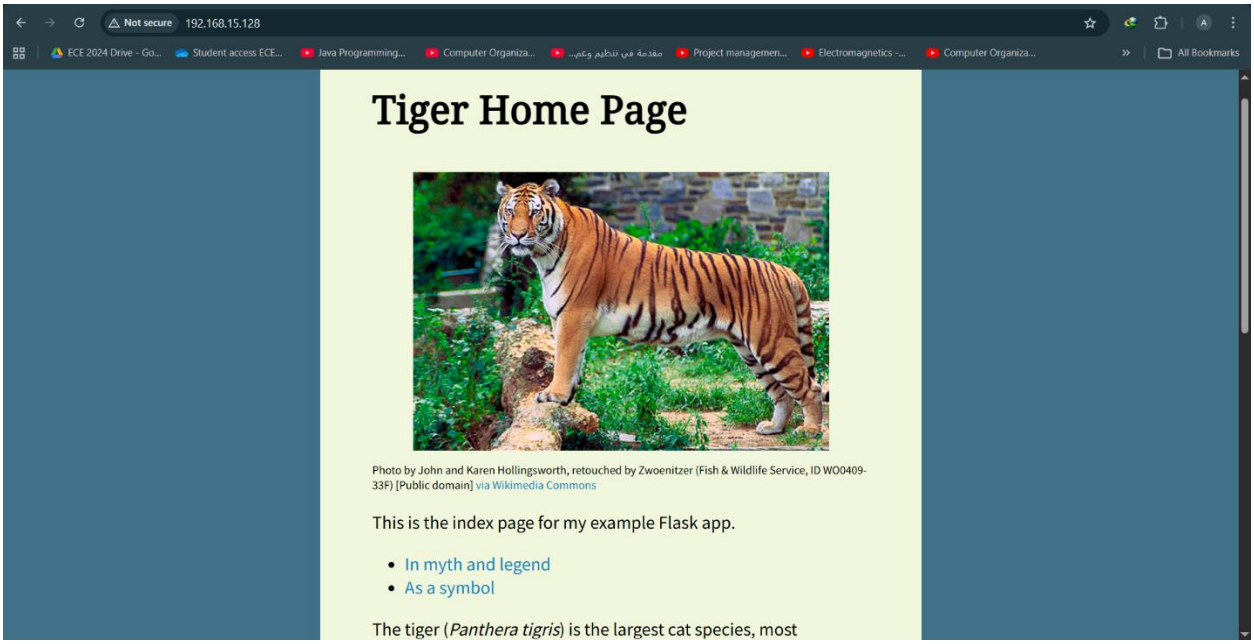
Step 4: Run Compose

docker compose up -d

```
WARN[0000] /root/flask-lab3/docker-compose.yml: the attribute `version` is
obsolete, it will be ignored, please remove it to avoid potential confusion

[+] up 3/3
 ✔ Network flask-lab3_default Created
                                                                    0.2s
 ✔ Container flask_app        Started
                                                                    0.3s
 ✔ Container nginx_proxy      Started
                                                                    0.9s
```

Step 5: Verify

```
[root@server flask-lab3]# curl http://localhost
<!DOCTYPE html>
<html>
<head>
        <meta charset="utf-8">
        <meta name="viewport" content="width=device-width, initial-scale=1"
>
        <title>Tiger Home Page</title>
  <link href="https://fonts.googleapis.com/css?family=Droid+Serif|Source+Sa
ns+Pro" rel="stylesheet">
        <link rel="stylesheet" href="/static/css/main.css">
</head>
```


Not secure
192.168.15.128
ECE 2024 Drive - Go...
Student access ECE...
Java Programming...
Computer Organiza...
مقدمة في تنظيم وعم...
Project managemen...
Electromagnetics -...
Computer Organiza...
All Bookmarks
Tiger Home Page
Photo by John and Karen Hollingsworth, retouched by Zwoenitzer (Fish & Wildlife Service, ID WO0409-33F) [Public domain] via Wikimedia Commons
This is the index page for my example Flask app.
In myth and legend
As a symbol
The tiger (Panthera tigris) is the largest cat species, most